University of Cape Town Honours in Statistics - Applied Spatial Data Analysis Elective Module

Dr Şebnem Er

Table of contents

Overview

ASDA is designed to introduce you to more advanced statistical techniques suitable for spatial datasets. The course will be oriented towards the analysis of spatial data. Applications will be drawn from a wide variety of fields in the natural, business and social sciences. Emphasis will also be placed on the interpretation of computer generated results.

Lectures

Tuesdays, 14:00–16:00 | **Wednesdays**, 09:00–11:00

Lecturers:

- **Dr Sebnem Er** (sebnem.er@uct.ac.za) 

Dr. Sebnem Er (she/her) is a senior lecturer at UCT, Statistical Sciences Department, and program coordinator for MSc Data Science degree at UCT.

I am located at PD Hahn Building 5th Floor, Room 5.55 however I will not be always in my office. If you would like to see me in person in my office please schedule a meeting by sending an email: Sebnem.Er@uct.ac.za

- **Mr Sulaiman Salau** (sulaiman.salau@uct.ac.za)

Mr. Sulaiman Salau (he/him) is a lecturer at UCT, Statistical Sciences Department at UCT, located at room 5.5x PD Hahn. Please email Mr Sulaiman Salau to make an appointment Sulaiman.Salau@uct.ac.za.

Consultation: By email appointment

Course Synopsis

Applied Spatial Data Analysis introduces statistical methods for analysing data in which geographical location plays an important role. The course focuses on practical implementation using **R** and covers the acquisition, management, visualisation, exploration, and modelling of spatial data.

Applications will be drawn from the natural, environmental, and social sciences, as well as public health, economics, and business. Emphasis is placed on interpreting statistical results and communicating findings effectively across three major classes of spatial data: **Geostatistical data**, **Point pattern data**, and **Areal (lattice) data**.

Learning Outcomes

By the end of the course, students should be able to:

- **Understand & Manage:** Distinguish spatial data types, understand coordinate reference systems/projections, and use R to import, manage, and transform spatial data.
- **Analyse & Explore:** Conduct exploratory spatial data analysis, identify spatial relationships, and analyse geostatistical, point pattern, and areal data using appropriate statistical methods.
- **Communicate & Reproduce:** Interpret results to draw sound conclusions, create informative maps/visualisations, and produce reproducible reports using R and Quarto.

Software & Materials

Data analysis forms a central part of this course. All demonstrations will use **R**. Students may use alternative software, but instruction and support will only be provided for R. Previous programming experience is helpful but not required.

Required Software: Ensure you have the latest versions of **R**, **RStudio**, and **Quarto** installed before the first practical. (Git is optional but recommended).

Primary References: (Lecture slides and practicals will be provided via Amathuba)

1. Bivand, R. et al. (2013). *Applied Spatial Data Analysis with R*. Springer. [https://asdar-book.org/]
2. Lovelace, R. et al. (2019). *Geocomputation with R*. CRC Press. [https://r.geocompx.org/]
3. Pebesma, E., & Bivand, R. (2023). *Spatial Data Science*. CRC Press. [https://r-spatial.org/book/]
4. Moraga, P. (2023). *Spatial Statistics for Data Science*. CRC Press. [https://www.paulamoraga.com/book-spatial/]

Assessment

! Duly Performed (DP) Requirement

- Students must obtain **at least 40%** for the class record in order to qualify for the final examination.
- There are no deferred exams at Honours.

Your final grade is composed of a **Class Record (60%)** and a **3-Hour Final Examination (40%)**.

Assessment Component	Weight	Description
Participation	15%	Active engagement in lectures, discussions, and Q&A sessions.
Presentation	15%	Presenting and critically discussing assigned readings.
Test 1	35%	Date to be announced.
Test 2	35%	Date to be announced.

$$\textbf{Final Mark} = \textbf{0.60} \times (\textbf{Class Record}) + \textbf{0.40} \times (\textbf{Examination Mark})$$

Course Policies

Communication & Changes Announcements, assessment information, and course material will be distributed through **Amathuba**. The lecturers reserve the right to amend this outline where necessary; students are responsible for monitoring announcements for updates.

Academic Honesty Students must maintain the highest standards of academic honesty. Submitted work must represent the student's own work unless collaboration is explicitly permitted. Familiarise yourself with UCT's plagiarism rules (https://uct.ac.za/administration/policies). Plagiarism will not be tolerated.

> Statement on the Use of AI
>
> While generative AI can provide useful tools, inappropriate AI use can undermine student learning, lead to academic dishonesty, and produce inaccurate or biased work.
> Therefore, **submitted work should not be produced using any generative AI tools** including using AI to identify/summarise readings, generate code structure, or write/edit text except with the express written permission of the module lecturer detailing appropriate use. Your task as a student is to develop core academic skills independent of AI.

Lecture Content

All content for this course is available on Amathuba and in this link. You will find all the necessary slides, pdfs, `R` examples in this link.